STAT 641 **Midterm Exam 2 Solution** **Ghosh**

1. Solution:

   (i) $\forall\ \omega \in (0,1], X_n(\omega) = 1$ for large n such that $0 < \frac{1}{n} < \omega < 1$. So $X_n(\omega) \to X(\omega), \forall\ \omega \in (0,1]$, and $P(\{0\}) = 0$ So. $X_n \to X$ a.e. (P).

   (ii) $m(\{\omega : |X_n(\omega) - X(\omega)| > \epsilon\}) = m([0, \frac{1}{n}]) = \frac{1}{n} \to 0$ as $n \to \infty$, $\forall\ \epsilon > 0$. So $X_n \xrightarrow{p} X$.

   (iii) Fix a $T > 1$. Then for each $n \geq 1$

   $$\int |X_n| I_{\{|X_n| > T\}} dP = (1+n)m([0, \frac{1}{n}]) = 1 + \frac{1}{n} > 1,$$

   So supremum of the integral (over n) is no smaller than 1 and hence doesnt go to zero when $T \to \infty$. So they are not UI.

   (iv) $(E|X_n - X|^p) = \left(\int |X_n - X|^p\ dP\right) = n^p \cdot m([0, \frac{1}{n}]) = n^{p-1} \to 0$ as $n \to \infty$ iff $0 < p < 1$. Hence $X_n \to X$ in L^p for $0 < p < \frac{1}{2}$, but this doesn't hold for $1 \leq p < \infty$.

2. Solution:

   (a) Define $p' = \frac{p}{r}, q' = \frac{q}{r}$, then $\frac{1}{p'} + \frac{1}{q'} = 1$ and hence, by Holder's inequality,

   $$E(|YZ|^r) = E(|Y^r Z^r|) \leq \left(E\left((Y^r)^{p'}\right)\right)^{\frac{1}{p'}} \left(E\left((Z^r)^{q'}\right)\right)^{\frac{1}{q'}} = \left(E\left(Y^p\right)\right)^{\frac{1}{p'}} \left(E\left(Z^q\right)\right)^{\frac{1}{q'}} < \infty.$$

   This is for $1 < p, q < \infty$. The result is true if either of them is ∞ or 1, but no need to prove it.

   (b) Take $\phi(x) = e^{x^2}$ - it is a convex function (easy to check). Using Jensen's inequality with this, we get $E(\phi(X)) \geq \phi(E(X))$, leading to the inequality.

3. Solutions:

(a) $\mu_3(\emptyset) = 3\mu_1(\emptyset) + 2\mu_2(\emptyset) = 0$; and $\mu_3(A) = 3\mu_1(A) + 2\mu_2(A) \geq 0 \Rightarrow \mu_3(A) \in [0, \infty]$, $\forall\ A \in \mathcal{F}$; If $\{A_i\}$ is a disjoint collection of sets in $\mathcal{F}$, then $\mu_3(\cup_i A_i) = 3\mu_1(\cup_i A_i) + 2\mu_2(\cup_i A_i) = 3\sum_i \mu_1(A_i) + 2\sum_i \mu_2(A_i) = \sum_i (3\mu_1(A_i) + 2\mu_2(A_i)) = \sum_i \mu_3(A_i)$.
So μ_3 is a measure.

(b) $\nu(A) = 0 \Rightarrow \mu_1(A) = \mu_2(A) = 0 \Rightarrow \mu_3(A) = 3\mu_1(A) + 2\mu_2(A) = 0$. So $\mu_3 \ll \nu$.
Will show: $\frac{d\mu_3}{d\nu} = 3\frac{d\mu_1}{d\nu} + 2\frac{d\mu_2}{d\nu}$ a.e (ν). Since $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, then $\exists\ f_1, f_2 \geq 0$ s.t $\mu_1(A) = \int_A f_1\ d\nu, \mu_2(A) = \int_A f_2\ d\nu$. By definition, $\mu_3(A) = 3\mu_1(A) + 2\mu_2(A) = 3\int_A f_1\ d\nu + 2\int_A f_2\ d\nu = \int_A (3f_1 + 2f_2)\ d\nu$ by linearity of integrals. Hence by uniqueness of RN derivatives, $\frac{d\mu_3}{d\nu} = 3f_1 + 2f_2 = 3\frac{d\mu_1}{d\nu} + 2\frac{d\mu_2}{d\nu}$ a.e (ν).

(c) Let $B = \mathbb{N} \cup \{0\} = \mathbb{N}_0$, then $3\mu_1(B^c) = 0 = \nu(B)$, i.e. $3\mu_1 \perp \nu$. Choose $\mu_a = 2\mu_2, \mu_s = 3\mu_1$, then $\mu = \mu_a + \mu_s$. To show $\mu_a \ll \nu$ and compute $\frac{d\mu_a}{d\nu}$.
Note that if $\phi_{1,1}$ represent the density of $Normal(1,1)$, then by definition $\mu_2(A) = \int_A \phi_{1,1} dm$ and $\nu = m$. So, $\frac{d\mu_2}{d\nu} = \phi_{1,1}$ from RN theorem. Hence, from the proposition in the text, $\frac{d\mu_a}{d\nu} = \frac{d(2\mu_1)}{d\nu} = 2\frac{d\mu_1}{d\nu} = 2\phi_{1,1}$.